

P₁

(a) False

(b) True

(c) False

(d) False

(e) False

NOTE: DNS lookup uses UDP, so there is only one RTT incurred with DNS servers.

P₇

$$\text{time elapsed} = \underbrace{RTT_0}_{\substack{\text{set-up} \\ \text{TCP connection}}} + \underbrace{RTT_1 + \dots + RTT_n}_{\substack{\text{Resolve name-} \\ \text{to-ip mapping}}} + \underbrace{RTT_0}_{\substack{\text{request} \Delta \\ \text{recieve} \\ \text{object.}}}$$

P₈

open connection for each object & recieve it

Resolve DNS

(a) time elapsed = $8(RTT_0 + RTT_0) + \overbrace{RTT_1 + \dots + RTT_n}^{\text{Resolve DNS}} + \underbrace{2RTT_0}_{\substack{\text{Recieve} \\ \text{base} \\ \text{html}}}$

(b) time elapsed = $\underbrace{(RTT_0 + RTT_0)}_{\substack{\text{Recieve 6} \\ \text{objects}}} + \underbrace{(RTT_0 + RTT_0)}_{\substack{\text{Recieve two} \\ \text{remaining objects}}} + RTT_1 + \dots + RTT_n + 2RTT_0$

(c) time elapsed = $\underbrace{RTT_0}_{\substack{\text{open} \\ \text{connection}}} + 8RTT_0 + RTT_1 + \dots + RTT_n + 2RTT_0$

P₉

Object size = 1000000 bits

Request Rate = 16 requests/sec

$$\text{Internet delay} = 3 \text{ sec}$$

$$\text{Access delay} = \frac{1000000 / 15000000}{1 - (1000000 / 15000000)(16)}$$

$$= 0.99 \text{ sec}$$

β : requests (objects) / sec

$$\text{Average response time} = 3 + 0.99 \text{ sec}$$

$$= 3.99 \text{ sec}$$

$$(b) \text{ Access delay} = \frac{1000000 / 15000000}{1 - (1000000 / 15000000)(16)(0.4)}$$

$$= 0.116 \text{ sec}$$

Requests that go to the origin server

$$\text{Average response time} = 0.4(3 + 0.116) + 0.6(0)$$

$$= 1.246 \text{ sec}$$

P_{100}

$$\text{link length} = 10 \text{ m } (L)$$

$$\text{Transmission rate} = 150 \text{ bits/sec } (R)$$

$$\text{data packet size} = 100,000 \text{ bits}$$

$$\text{Control size} = 200 \text{ bits}$$

$$\text{Object size} = 100 \text{ kbits}$$

$$\text{As each object will get } \frac{150 \text{ bit/sec}}{10} = 15 \text{ bits/sec.}$$

So, transmission delay will be:

$$\frac{100,000}{15} = 6,666.67 \text{ secs.}$$

For persistent case,

$$\begin{aligned}\text{Transmission delay for one packets} &= \frac{100000 \text{ bits} \times 10}{150 \text{ bits/sec}} \\ &= 666.67 \text{ sec.}\end{aligned}$$

So,

$$\begin{aligned}\text{Total} &= 11 \times 666.67 \\ &= 7333.37 \text{ sec.}\end{aligned}$$

So, there is no significant gain.

Interactive end of chapter Exercises

DNS & HTTP delays

1.

$$2RTT_{\text{HTTP}} + RTT_0 + RTT_1 + RTT_2$$

$$2(24) + 2 + 27 + 50 \text{ msec}$$

$$127 \text{ ms}$$

$$2. 10(2RTT_{\text{HTTP}}) + 2RTT_{\text{HTTP}} + RTT_0 + RTT_1 + RTT_2$$

$$10(2 \times 24) + 127$$

$$607 \text{ msec}$$

$$3. 2RTT_{\text{HTTP}} + 2RTT_{\text{HTTP}} + 2RTT_{\text{HTTP}} + RTT_0 + RTT_1 + RTT_2$$

$$2(24) + 2(24) + 127$$

$$223 \text{ ms}$$

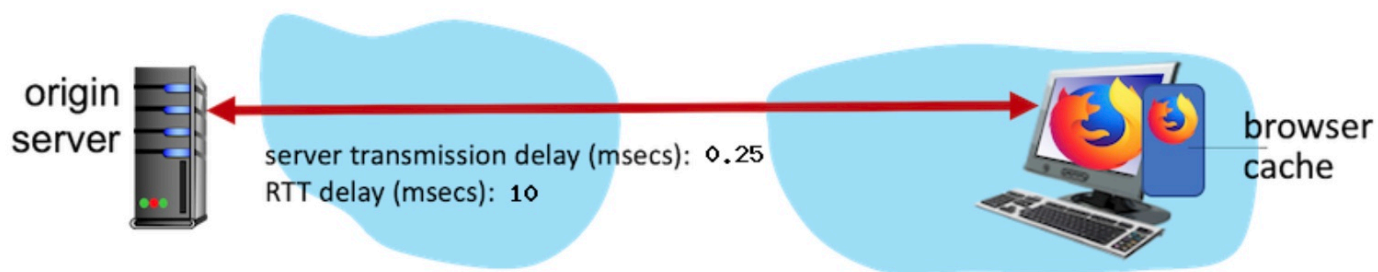
$$3. \quad RTT_{\text{HTTP}} + RTT_{\text{HTTP}} + 2RTT_{\text{HTTP}} + RTT_0 + RTT_1 + RTT_2$$

$$24 + 24 + 127$$

$$175 \text{ms}$$

Browser Caching

Consider an HTTP server and client as shown in the figure below. Suppose that the RTT delay between the client and server is 10 msec; the time a server needs to transmit an object into its outgoing link is 0.25 msec; and any other HTTP message not containing an object has a negligible (zero) transmission time. Suppose the client again makes 60 requests, one after the other, waiting for a reply to a request before sending the next request.



Assume the client is using HTTP 1.1 and the IF-MODIFIED-SINCE header line. Assume 30% of the objects requested have NOT changed since the client downloaded them (before these 60 downloads are performed)

QUESTION 1 OF 1

How much time elapses (in milliseconds) between the client transmitting the first request, and the completion of the last request?

$$\text{Hit rate} = 0.3$$

$$\text{Requests catered by the server} = 60 \times 0.7$$

$$= 42 \text{ requests}$$

$$\text{Delay} = RTT + \underbrace{60 \times RTT}_{\substack{\text{All request are made those who aren't modified} \\ \text{incurs 0 Trans}}} + 42 \times 0.25$$

$$= 620.5 \text{ sec}$$